

Lab07

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
path <- "."  
fname <- file.path(path, "uwmadison.sqlite3")  
conn = dbConnect(SQLite(), fname)
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

Código EXTRA usado exclusivamente para auxiliar a identificar colunas a serem usadas no JO

```
tabelas <- dbListTables(conn)  
  
for (tabela in tabelas) {  
  cat("\nTabela:", tabela, "\n")  
  print(dbListFields(conn, tabela))  
}
```

```
Tabela: course_offerings  
[1] "uuid"          "course_uuid" "term_code"   "name"
```

```
Tabela: courses  
[1] "uuid"  "name"  "number"
```

```
Tabela: grade_distributions
```

[1]	"course_offering_uuid"	"section_number"	"n_count"
[4]	"c_count"	"f_count"	"cr_count"
[7]	"p_count"	"nw_count"	"other_count"
[10]	"bc_count"	"u_count"	"d_count"
[13]	"ab_count"	"a_count"	"s_count"
[16]	"i_count"	"b_count"	"nr_count"

Tabela: instructors

[1]	"id"	"name"
-----	------	--------

Tabela: rooms

[1]	"uuid"	"facility_code"	"room_code"
-----	--------	-----------------	-------------

Tabela: schedules

[1]	"uuid"	"start_time"	"end_time"	"mon"	"tues"
[6]	"wed"	"thurs"	"fri"	"sat"	"sun"

Tabela: sections

[1]	"uuid"	"course_offering_uuid"	"section_type"
[4]	"number"	"room_uuid"	"schedule_uuid"

Tabela: subject_memberships

[1]	"subject_code"	"course_offering_uuid"
-----	----------------	------------------------

Tabela: subjects

[1]	"code"	"name"	"abbreviation"
-----	--------	--------	----------------

Tabela: teachings

[1]	"instructor_id"	"section_uuid"
-----	-----------------	----------------

```
professores_stat = dbGetQuery(conn,
  'SELECT instructors.name, subjects.abbreviation FROM instructors
  JOIN teachings
  ON instructors.id = teachings.instructor_id
  JOIN sections
  ON teachings.section_uuid = sections.uuid
  JOIN subject_memberships
  ON sections.course_offering_uuid = subject_memberships.course_offering_uuid
  JOIN subjects
  ON subject_memberships.subject_code = subjects.code
  WHERE subjects.abbreviation = "STAT"
  GROUP BY instructors.name
```

')

professores_stat

	name	abbreviation
1	ABIGAIL BENZINE	STAT
2	ADIN-CRISTIAN ANDREI	STAT
3	AKICHIKA OZEKI	STAT
4	ALAN HUANG	STAT
5	ALBERTO DEL PIA	STAT
6	ALBRECHT KLEMM	STAT
7	ALEXANDER COVINGTON	STAT
8	ALEXANDER FISH	STAT
9	ALEXANDER KISELEV	STAT
10	ALYSSA DIGILIO	STAT
11	ANATOLE BECK	STAT
12	ANDREJ ZLATOS	STAT
13	ANDREW LESLIE	STAT
14	ANNE BRUCKNER	STAT
15	ANQI SHI	STAT
16	ANRU ZHANG	STAT
17	ARNOLD MILLER	STAT
18	AUGUST JENSEN	STAT
19	BEHZAD AALIPUR	STAT
20	BEHZAD AALIPUR HAFSHEJANI	STAT
21	BEN ADAM HAALAND	STAT
22	BENEDEK VALKO	STAT
23	BENJAMIN RECHT	STAT
24	BI CHENG WU	STAT
25	BIN DAI	STAT
26	BIN ZHANG	STAT
27	BIN ZHU	STAT
28	BINGYING XIE	STAT
29	BO HUANG	STAT
30	BO YANG	STAT
31	BOWEN HU	STAT
32	BRET HANLON	STAT
33	BRET LARGET	STAT
34	BRIAN YANDELL	STAT
35	BROOK LUERS	STAT
36	CECILE ANE	STAT
37	CHAN PARK	STAT

38	CHANHAN HSU	STAT
39	CHAOQUN MEI	STAT
40	CHAOYANG YU	STAT
41	CHELSEY GREEN	STAT
42	CHEN CHENG	STAT
43	CHEN JING	STAT
44	CHENGNING ZHANG	STAT
45	CHENLIANG XU	STAT
46	CHENSHENG KUANG	STAT
47	CHENXI LI	STAT
48	CHIA-CHIEH LIN	STAT
49	CHIEN-WEI CHEN	STAT
50	CHRISTINA M. KENDZIORSKI	STAT
51	CHRISTINE SORKNESS	STAT
52	CHRISTOPHER WAGNER	STAT
53	CLAIRE BOBST	STAT
54	CLAUDIA SOLIS LEMUS	STAT
55	COLE COOK	STAT
56	COLIN LONGHURST	STAT
57	CRYSTAL CHEN	STAT
58	CUICUI QI	STAT
59	CUIZE HAN	STAT
60	DANIELE CAPPELLETTI	STAT
61	DAVID ANDERSON	STAT
62	DAVID DEMETS	STAT
63	DAVID GRIFFEATH	STAT
64	DEBRAJ DAS	STAT
65	DEREK BEAN	STAT
66	DEREK NORTON	STAT
67	DEYUAN JIANG	STAT
68	DIETRICH UHLENBROCK	STAT
69	DONALD PORTER	STAT
70	DONG XIA	STAT
71	DONGGYU KIM	STAT
72	DONGHYUN LEE	STAT
73	DOUGLAS M. BATES	STAT
74	DUY NGUYEN	STAT
75	DUZHE WANG	STAT
76	EDWARD ERKER	STAT
77	ELOISA D CHAVAS	STAT
78	ERICA LEE DEADMAN	STAT
79	ERIK NORDHEIM	STAT
80	FAN GAO	STAT

81	FAN YANG	STAT
82	FANG FANG	STAT
83	FANGFANG WANG	STAT
84	FLORIAN BERTRAND	STAT
85	FREDERICK BOEHM	STAT
86	GARVESH RASKUTTI	STAT
87	GARY HOWARD SCHROEDER	STAT
88	GENG LI	STAT
89	GINA BENNINGER	STAT
90	GINA OH	STAT
91	GONZALO CONTADOR	STAT
92	GRACE WAHBA	STAT
93	GREGORIO MORENO-FLORES	STAT
94	GREGORY SHINAULT	STAT
95	GUANHUA CHEN	STAT
96	GUANNAN SUN	STAT
97	GUILHERME ROSA	STAT
98	GUILHERME VIEIRA NUNES LUDWIG	STAT
99	GUN WOONG PARK	STAT
100	HAN CHEN	STAT
101	HAO CHEN	STAT
102	HAO TENG	STAT
103	HAO ZHENG	STAT
104	HAO ZHOU	STAT
105	HAODA FU	STAT
106	HAOYANG FAN	STAT
107	HEATHER MARIE BRAZEAU	STAT
108	HUAIBAO FENG	STAT
109	HUI WANG	STAT
110	HUIKUN ZHANG	STAT
111	HUILIN HU	STAT
112	HYEBIN SONG	STAT
113	HYUNSEUNG KANG	STAT
114	IAN BRANSTAD RILEY	STAT
115	ISMOR FISCHER	STAT
116	JAMES ANDERSON	STAT
117	JAMES D KUELBS	STAT
118	JARED BROWN	STAT
119	JASON P FINE	STAT
120	JASON RICHARD SWANSON	STAT
121	JEE YEON KIM	STAT
122	JEEA CHOI	STAT
123	JENNIFER BIRSTLER	STAT

124	JENNIFER NGUYEN	STAT
125	JESSE THOMAS HOLZER	STAT
126	JIAJIE CHEN	STAT
127	JIALE XU	STAT
128	JIAN WU	STAT
129	JIANCHANG HU	STAT
130	JIE SONG	STAT
131	JIE WEI	STAT
132	JIE ZHANG	STAT
133	JILI WANG	STAT
134	JINGCI MENG	STAT
135	JINGJIANG PENG	STAT
136	JINGLAN LI	STAT
137	JIWEI ZHAO	STAT
138	JOHN DAVIS	STAT
139	JOHN GILLETT	STAT
140	JOHN KANE	STAT
141	JOHN WILTSHIRE-GORDON	STAT
142	JONATHON PETERSON	STAT
143	JOSEP GINEBRA	STAT
144	JOSEPH DEUTSCH	STAT
145	JU HEE CHO	STAT
146	JUN LI	STAT
147	JUN SHAO	STAT
148	JUN YIN	STAT
149	JUN ZHANG	STAT
150	JUN ZHU	STAT
151	JUNGWON MUN	STAT
152	JUNHEE HAN	STAT
153	JUNHO LEE	STAT
154	KAILEI CHEN	STAT
155	KAM-WAH TSUI	STAT
156	KARL BROMAN	STAT
157	KARL ROHE	STAT
158	KATHERINE GOODE	STAT
159	KAZUHIKO SHINKI	STAT
160	KEEGAN KORTHAUER	STAT
161	KEN ONO	STAT
162	KEVIN HASEGAWA ENG	STAT
163	KEVIN PACKARD	STAT
164	KJELL DOKSUM	STAT
165	KRISHNAKUMAR BALASUBRAMANIAN	STAT
166	KRISTEN CYFFKA	STAT

167	KUNLING HUANG	STAT
168	KYLE HEBERT	STAT
169	KYLE HERBET	STAT
170	KYUNGMAHN KIM	STAT
171	LAM HO	STAT
172	LAN LUO	STAT
173	LANCINE KONATE	STAT
174	LEI XU	STAT
175	LEV BORISOV	STAT
176	LIAM JOHNSTON	STAT
177	LIE XIONG	STAT
178	LILI LAN	STAT
179	LILI ZHENG	STAT
180	LILUN DU	STAT
181	LIN QI	STAT
182	LONG PHAN	STAT
183	LU MAO	STAT
184	LU YANG	STAT
185	LUWAN ZHANG	STAT
186	LUXI CAO	STAT
187	MANJUSHA KANCHARLA	STAT
188	MARI PALTA	STAT
189	MARIA KAMENETSKY	STAT
190	MARIA ROJO	STAT
191	MARIAN R FISHER	STAT
192	MARY LINDSTROM	STAT
193	MATTHEW BALLARD	STAT
194	MENG SONG	STAT
195	MICHAEL FERRIS	STAT
196	MICHAEL GEORGE ILLIS	STAT
197	MICHAEL HOGAN	STAT
198	MICHAEL KUTZLER	STAT
199	MICHAEL LIOU	STAT
200	MICHAEL NEWTON	STAT
201	MICHAEL RENE KOSOROK	STAT
202	MICHELLE HARRIS	STAT
203	MIHAELA IFRIM	STAT
204	MIN JUNG LEE	STAT
205	MIN NIU	STAT
206	MING XIE	STAT
207	MING YUAN	STAT
208	MINJING TAO	STAT
209	MITCHELL PAUKNER	STAT

210	MOO K CHUNG	STAT
211	MUHONG GAO	STAT
212	MURRAY CLAYTON	STAT
213	NATALIA DE LEON GATTI	STAT
214	NELLIE LAUGHLIN	STAT
215	NICHOLAS HENDERSON	STAT
216	NICHOLAS STEPHEN KEULER	STAT
217	NING FAN	STAT
218	NORBERT BINKIEWICZ	STAT
219	NORMAN DRAPER	STAT
220	PAUL M TERWILLIGER	STAT
221	PAUL RATHOUZ	STAT
222	PAUL SAVARIAPPAN	STAT
223	PEIGEN ZHOU	STAT
224	PERLA REYES	STAT
225	PHILIP WOOD	STAT
226	QI JIANG	STAT
227	QI TANG	STAT
228	QIAN ZHIGUANG	STAT
229	QING LI	STAT
230	QIONG ZHANG	STAT
231	QIURONG CUI	STAT
232	QUEFENG LI	STAT
233	QUOC TRAN	STAT
234	REBECCA KOSCIK	STAT
235	REBECCA POST	STAT
236	RICHARD A BRUALDI	STAT
237	RICHARD A. JOHNSON	STAT
238	RICHARD J. CHAPPELL	STAT
239	ROBERT HARRON	STAT
240	ROBERT R MEYER	STAT
241	ROBERT WARDROP	STAT
242	ROBERT WAYNE GREEN	STAT
243	RONALD GANGNON	STAT
244	RONGJUN ZHU	STAT
245	RUI CHEN	STAT
246	RUI TANG	STAT
247	RUIFANG SONG	STAT
248	RUIFENG XU	STAT
249	RUNGANG HAN	STAT
250	RUOSI GUO	STAT
251	RYAN ZEA	STAT
252	SAMUEL STECHMANN	STAT

253	SANGBUM CHOI	STAT
254	SCOTT HOTTOVY	STAT
255	SCOTT JOSEPH HETZEL	STAT
256	SEAN KENT	STAT
257	SEBASTIEN ROCH	STAT
258	SEHO PARK	STAT
259	SEULKEE YUN	STAT
260	SEUNGBONG HAN	STAT
261	SHAN LU	STAT
262	SHANE HUBLER	STAT
263	SHANG WU	STAT
264	SHENG WANG	STAT
265	SHENG ZHANG	STAT
266	SHENGJI JIA	STAT
267	SHI JIN	STAT
268	SHIXUE LIU	STAT
269	SHIZHEN WANG	STAT
270	SHUAI CHEN	STAT
271	SHUANG HUANG	STAT
272	SHULEI WANG	STAT
273	SHUYUN YE	STAT
274	SIJIAN WANG	STAT
275	SIJING LI	STAT
276	SOHEIL SADEGHI	STAT
277	SOKOL VAKO	STAT
278	SONA Z SWANSON	STAT
279	SONG WANG	STAT
280	SRIKANTHMADHAVAN ARAVAMUTHAN	STAT
281	STEFFEN LEMPP	STAT
282	STEPHEN AARON STANHOPE	STAT
283	STEPHEN BERG	STAT
284	STEPHEN WAINGER	STAT
285	STEPHEN WRIGHT	STAT
286	STEVEN SAM	STAT
287	SUKHENDU MEHROTRA	STAT
288	SUNDUZ KELES	STAT
289	TAERI UHM	STAT
290	TAO YU	STAT
291	THEVAASIINEN CHANDERENG	STAT
292	THOMAS COOK	STAT
293	THOMAS G. KURTZ	STAT
294	THU LE	STAT
295	TIEN VO	STAT

296	TIMO SEPPALAINEN	STAT
297	TIMOTHY IDOWU	STAT
298	TING YE	STAT
299	TING-LI LIN	STAT
300	TONG LI	STAT
301	TONGHAI YANG	STAT
302	TRAM TA	STAT
303	TUN LEE NG	STAT
304	TZU HSIANG HUNG	STAT
305	VARSHA KULKARNI	STAT
306	VICTOR LUO	STAT
307	VICTORIA MANSFIELD	STAT
308	WAI TONG FAN	STAT
309	WEI ZHENG	STAT
310	WEI-YIN LOH	STAT
311	WENWEN ZHANG	STAT
312	WENZHI CAO	STAT
313	WESLEY CHANG	STAT
314	XIAO GUO	STAT
315	XIAO NIE	STAT
316	XIAODAN WEI	STAT
317	XIAOMAO LI	STAT
318	XIAOPING FENG	STAT
319	XIAOWU DAI	STAT
320	XIN LI	STAT
321	XIN ZHANG	STAT
322	XINJIE HE	STAT
323	XINWEI DENG	STAT
324	XINXIN YU	STAT
325	XINYU SONG	STAT
326	XIPEI YANG	STAT
327	XIRAN WANG	STAT
328	XIUFENG SHAO	STAT
329	XIUYU MA	STAT
330	XIWEN MA	STAT
331	XU HE	STAT
332	XU XU	STAT
333	XUEYAO CHEN	STAT
334	XUN ZHAO	STAT
335	YAJUAN SI	STAT
336	YALI WANG	STAT
337	YAN CHEN	STAT
338	YANBING ZHENG	STAT

339	YANG ZHAO	STAT
340	YANNAN QIU	STAT
341	YAOGUO XIE	STAT
342	YAoyao XU	STAT
343	YAZHEN WANG	STAT
344	YI CHAI	STAT
345	YI LI	STAT
346	YI LIU	STAT
347	YIFAN MEI	STAT
348	YILIN ZHANG	STAT
349	YING ZHANG	STAT
350	YONGFENG WU	STAT
351	YONGJOON KIM	STAT
352	YONGSU LEE	STAT
353	YOUNG LEE	STAT
354	YOUNG MIN PARK	STAT
355	YOUNGDEOK HWANG	STAT
356	YOURAN QI	STAT
357	YU MENGANG	STAT
358	YU QIU LIU	STAT
359	YUAN JIANG	STAT
360	YUAN LI	STAT
361	YUAN WANG	STAT
362	YUANZHI LI	STAT
363	YUCHANG WU	STAT
364	YUCHEN ZHOU	STAT
365	YUJIN CHUNG	STAT
366	YUNONG LIN	STAT
367	YUQING XU	STAT
368	ZHANG CHUNMING	STAT
369	ZHENGJUN ZHANG	STAT
370	ZHENGXIAO WU	STAT
371	ZHIGENG GENG	STAT
372	ZHIGUO XIAO	STAT
373	ZHONGJIE YU	STAT
374	ZHUANG WU	STAT
375	ZIFENG ZHAO	STAT
376	ZIJIAN NI	STAT
377	ZUOFENG SHANG	STAT

```
dim(professores_stat)
```

```
[1] 377    2
```

3. O GPA americano é definido numa escala de 0 a 4, em que $A = 4$, $AB = 3.5$, $B = 3$, $BC = 2.5$, $C = 2$, $D = 1$ e $F = 0$. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
GPA_professores <- dbGetQuery(conn, 'SELECT instructors.name as professor, subjects.name as d
4*a_count +
3.5*ab_count +
3*b_count + 2.5*bc_count
+ 2*c_count +
d_count)/ (
  a_count +
  ab_count +
  b_count +
  bc_count +
  c_count +
  d_count +
  f_count) AS GPA
FROM instructors

JOIN teachings
  ON instructors.id = teachings.instructor_id

JOIN sections
  ON teachings.section_uuid = sections.uuid

JOIN grade_distributions
  ON sections.number = grade_distributions.section_number
  AND sections.course_offering_uuid = grade_distributions.course_offering_uuid

JOIN subject_memberships
  ON grade_distributions.course_offering_uuid = subject_memberships.course_offering_u

JOIN subjects
  ON subject_memberships.subject_code = subjects.code

WHERE subjects.abbreviation = "STAT"
GROUP BY instructors.name')
```

- Quem é o professor mais difícil?
- Quem é o professor mais fácil?
- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

```
library(tidyverse)
```

Warning: pacote 'ggplot2' foi compilado no R versão 4.5.3

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Warning: pacote 'forcats' foi compilado no R versão 4.5.3

Warning: pacote 'lubridate' foi compilado no R versão 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.1.5
```

```
v forcats   1.0.1      v stringr    1.5.1
```

```
v ggplot2    4.0.3      v tibble     3.2.1
```

```
v lubridate  1.9.5      v tidyr      1.3.1
```

```
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
professor_facil <- GPA_professores %>%
  filter(GPA == max(GPA, na.rm = TRUE))
```

```
professor_dificil <- GPA_professores %>%
  filter(GPA == min(GPA, na.rm = TRUE))
```

```
professor_facil
```

	professor	disciplina	GPA
1	GUANHUA CHEN	Statistics	4
2	SRIKANTHMADHAVAN ARAVAMUTHAN	Statistics	4

```
professor_dificil
```

```
      professor disciplina      GPA  
1 NICHOLAS STEPHEN KEULER Statistics 2.513158
```

```
# Só existe uma única disciplina de Estatística = 'Statistics', Por tanto ela vai ser a mais d
```

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-22 19:23:12 -03"
```